

Appendix 3C: Responses to CIE recommendations for the Alaska sablefish assessment

3C.1 Introduction

This document is a point by point response to specific recommendations raised in the panel summary of the 2009 Alaska sablefish assessment review. Recommendations or specific criticisms by the review panel are in **bold**. Our responses are in *italics*.

Terms of reference (TOR)

TOR 1: Evaluation, findings, and recommendations on quality of input data and methods used to process them for inclusion in the assessment.

TOR 2: Evaluation, findings, and recommendations on the level and adequacy of knowledge and incorporation of life history, ecology and habitat requirements.

TOR 3: Evaluation, findings, and recommendations of the analytical approach used to assess stock condition and stock status.

TOR 4: Evaluation, findings, and recommendations of areal apportionment of harvest strategy as related to optimizing spawning stock biomass.

TOR 5: Recommendations for further improvements.

3C.2 TOR 1: Input data and processing

Station locations. The panel noted that many of the locations for stations used to compute the survey index were originally selected by the Japanese fishing masters using knowledge of the spatial distribution of sablefish to set in areas with the highest possible catch rates whilst spreading them out as much as possible along the coast. This selection of stations could lead to bias in the survey index. The additional stations in gullies, which are not used in the assessment, could provide a useful check on this. **An analysis to check for differences in trends between the gully and non-gully stations may be a useful analysis to evaluate possible bias in the survey index.**

While the Japanese fishing masters may have located many stations in areas of high sablefish abundance, the survey does cover depths from 100 to 1,000 meters which includes a large range of sub-optimum sablefish depths. A comparison of gully versus standard survey stations will be presented in the 2009 SAFE report.

GOA bottom trawl survey. The Panel concluded that the inclusion of the GOA bottom trawl survey (to 500m) is potentially useful for indices of abundance for incoming year classes; and although the survey should help estimate recent year class strength, **the current assessment does not fit this index as well as the longline survey indices.**

While the trawl survey index generally supports a similar trend as the longline survey indices, the amount of weight given it in the model is small compared to the longline survey because of the use of both RPNs and RPWs and because it is assigned a higher annual sampling variability. Since the trawl survey only covers the depth range of younger sablefish, it is an index of year-class strength and we expect it to be variable and more difficult to fit closely. This issue should be alleviated when we use RPNs only for the 2009 assessment (see assessment section below).

Commercial longline fishery catch rates. The Panel concluded that the use of fishery CPUE is appropriate for the current assessment. **The practice of postscreening fishing operations to derive target-specific effort may lead to unwanted bias in the CPUE indices, and the Panel suggested that a better approach to evaluating the fishery CPUE would be to undertake a statistical (GLM) analysis.**

We are currently evaluating the use of Generalized Linear Models (GLM) to account for variables that may affect fishery CPUE. To avoid bias, sets will not be screened by deriving a target. This will not be finished for the 2009 assessment cycle, but will likely be incorporated in the 2010 assessment.

Accuracy of landings/total catch and stock structure. The Panel considers the current treatment of stock area and total catches as adequate for assessment and associated management. Nevertheless, the following points were noted:

State catches are not included in the assessment but their exclusion is unlikely to have a significant effect on the assessment results.

With the implementation of Annual Catch Limits (ACLs) in the coming assessment cycles, all state and research catches will be included in the catch estimates.

Catches from the western Bering Sea in the earlier part of the time period are unknown, and the overall catch figures for the earlier period when the fishery was open to international fleets is likely to be generally of poorer quality than in later years. The likely effect of underestimated catch on the assessment results is not quantifiable but is unlikely to have any significant effect on the recent stock biomass estimates. There is anecdotal information of high-grading during

different years. **The sensitivity of the assessment to alternative plausible catch history has not been investigated.**

Multiple attempts at alternative catch histories have been modeled throughout the history of the sablefish assessment to account for unknown Bering Sea catches, larger reported fish market reports in Japan than reported catch, and high-grading (Sigler et al. 1995, Sigler et al. 1999, etc). The current catch history is considered the most reasonable given the data available.

Age-length sampling. The Panel noted that the adequacy of length-age sampling has improved in recent years. Vessels accounting for 30% of the catch are sampled, which is relatively good coverage and indicates that the effective sample size is high. Trawl fishery data are sparse, but the longline fishery was well sampled. **The adequacy of existing sample size in terms of precision should be investigated.**

The tradeoffs between actual sample size and effective sample size will be an important part of upcoming modeling efforts.

The age-length conversion matrix appears to be appropriate. It was noted that the change in growth is modeled with a step-change. **An improvement may be to have a gradual change over a number of years.**

We agree that the abrupt growth change may induce unwanted residual patterns. In the future we will look at a smooth transition between growth regimes or consider estimating growth within the model.

Voluntary logbook scheme. The voluntary logbook program was seen as helpful to evaluate the under-60' fleet, which is otherwise only monitored based on fish-ticket data. Some concerns were raised that the coverage for this fleet was very low historically. **The implications of this low sampling level for this fleet component on the derived abundance index should be investigated.**

Voluntary logs from under-60 ft vessels are indistinguishable from required logs of vessels over 60 ft because of confidentiality agreements between the data collecting agency (the International Pacific Halibut Commission, IPHC) and the vessels. Proportions of voluntary and required logs are reported in the SAFE. However, because of our inability to separate the log data by vessel size, an analysis of the effects of increased coverage of the under-60 ft fleet is not possible. However, in the aforementioned standardization of observed fishery data using GLM's, vessel size will be analyzed.

Data not currently included in current assessment. A number of data sources not currently used in the assessment were identified as candidates for inclusion in future assessments and their utility should be investigated:

Combined sex data from early fishery size composition data

We may include these data in the 2009 assessment cycle in order to estimate selectivity for the early Japanese fisheries.

Sex ratio data can potentially be used in fitting the model

We have made some effort to include sex ratio data in the past, but are still working on a consistent method. Since we have compositional data from a number of different data sources that span different time periods, it is difficult to fit a time series of sex ratios. We agree that the implied sex ratio is potentially important to model results.

The time-series of sablefish CPUE from IPHC surveys

We have briefly investigated sablefish CPUE data from the IPHC commission. A table of the time series, for years when they were collecting relatively consistent sablefish data, will be included in the assessment document.

EBS slope surveys (although there are concerns regarding the sex ratio and a predominance of large males need to be investigated)

We will monitor the time series of EBS slope surveys. At this time the series is quite short, but a table will be included in the assessment document.

State surveys (recognizing potential issues with applicability to the AK-wide stock)

It is highly unlikely we would be able to use state surveys, which are regionalized, in the Alaska-wide assessment. If we move toward a spatially-explicit model in the future, we will investigate the utility of regionalized surveys.

3C.3 TOR 2: Life history, ecology, and habitat requirements

Maturity. The use of separate maturity ogives for female and male sablefish represents the most appropriate use of maturity data for computing spawning biomass rather than the use of a combined-sex maturity ogive. The Panel notes that the ogives currently used are from data collected prior to the mid-1980s, and that more recently collected and histologically verified maturity data are available and should be used in future assessments. The new data indicate a slightly higher age at 50% maturity in females. **Temporal trends in maturity should be monitored. However, given the observed changes in growth, it would be valuable to quantify the age and length dependence of maturation.**

We agree new age at 50% maturity estimates are needed but have not been able to acquire funding to conduct these studies. AFSC scientists are pursuing maturity sampling during the winter, weeks or

months before sablefish likely spawn in Alaska. Samples from this time of year are necessary for definitive maturity staging for use in age-at-maturity analyses. Investigations of using gonad samples from late in the IFQ fishing season (October-November) for age-at-maturity analyses are also in progress.

Ecosystem aspects and competition/predation levels that potentially impact sablefish stocks. The Panel supports efforts to quantify ecosystem effects on sablefish dynamics. **In particular, studies on factors affecting conditions for pre-recruits would be useful to provide insights on medium-term future trends. Such studies would benefit from reliable data on abundance trends for young sablefish from suitable surveys.** Large changes in predator/competitor species (e.g. the recent substantial increase in arrowtooth flounder abundance) may affect population trends of sablefish.

We agree. Several studies are underway looking at recruitment dynamics of sablefish in relation to climate forcing in Alaska. In addition the recent funding of the large Gulf of Alaska Integrated Ecosystem Research Plan has a major focus on recruitment dynamics of sablefish for the next several years.

3C.4 TOR 3: Analytical Approach

The Panel concludes that the analytical approach was appropriate and provides an acceptable basis for management advice. There was some double-use of longline relative indices (i.e. both RPNs and RPWs). Sensitivity analyses and diagnostics were requested and examined during the review.

Abundance indices. It was recommended that the RPN versions of the Japanese and domestic longline indices should be used and RPW values should be omitted since these indices are highly correlated. Use of both number and weight effectively doubles the weight given to these data in model fitting.

The current authors are in agreement with the CIE's thoughts on the double-use of this index and its potential to mask other data sources. We intend to remove the RPWs for the 2009 assessment cycle.

Retrospective pattern. The current assessment has a retrospective bias where successive assessments revise the entire biomass series downwards, with the largest bias occurring in the recent period up to 2006. The bias appears much reduced over the last two years of the assessment. The causes of this bias require further investigation, particularly in relation to the appropriateness of the current model configuration. **The impact of the bias on ABC estimates is uncertain and also warrants further investigation.**

We will continue to investigate the retrospective pattern in light of proposed changes to the model to see if some potential model misspecification has been alleviated.

Diagnostics and sensitivity analysis. The Panel requested some standard

assessment diagnostics and sensitivity analyses:

A comparison of input and output CVs for abundance indices. The base model configuration tended to produce larger output CVs than input CVs. **Results suggest that the CVs for all indices in the base model should be doubled.**

During the process of removing the RPW indices we will attempt to adjust the overall weights in the model to more accurately reflect the underlying uncertainty in each data source.

A comparison of input and effective sample sizes for compositional data. These indicate that the input N may be overestimated for the cooperative and domestic longline survey age data, and underestimated for other compositional data.

As mentioned above, as we remove the RPW indices it will require us to consider the relative input Ns in the compositional sources. We will include plots/tables of effective N and deviation of the normalized residuals (SDNR) in future assessments.

3C.5 TOR 4: Areal apportionment (and whales)

Use of survey indices and fishery CPUE data. The apportionment scheme provides more weight to the longline survey data for regional abundance than to the fishery CPUE data. The Panel agrees that, given the data available, this is appropriate, even given factors such as trends in fishery interaction and whale depredation with the survey. Although the longline survey covers only part of the fishing season, whilst the fishery CPUE data arises from information over the full 8-month season, the survey has the advantage of using a standardized design over the full area. Variation between areas and times in the fishery CPUE data may not fully reflect the pattern of abundance of sablefish due to targeting and differences in fishing gears. **The Panel recommends the use of region-specific selectivity/availability estimates be explored as a possible modification to the apportionment scheme.** This may lead to better use of the fishery and survey data for apportionment. Projections taking such selectivity factors into account could be used to evaluate the performance of different apportionment strategies.

We have previously explored simple methods such as apportioning based on the amount of fish at 50% selection or 50% maturity. While such schemes are appealing we generally contend that within a reasonable range, the apportionment scheme is not crucial biologically. We believe that socioeconomic objectives need to be clarified before any new apportionment scheme is devised. We are also directed by the SSC and Council process to use the current scheme. Because of the socioeconomic objectives, we feel any changes must be endorsed by the SSC and Council unless the apportionment scheme becomes a biological concern.

Tagging model. The Panel noted that movement estimates using results from the updated tagging model should be used for evaluating the impact of different apportionment schemes.

Updated results from tagging data appear to show that movement patterns have changed. This data will be taken into consideration in the design of any new apportionment scheme.

Whale depredation. The various impacts of whale depredation were examined. This was identified as potentially affecting the abundance index and the regional apportionments. The removal of killer whale depredated sections of sets from the longline survey index does not appear to create a bias but is likely to add to the uncertainty (i.e., higher variance due to smaller sampling effort). Killer whale depredation appears to be relatively stable over time. **Despite these observations there may be merit in evaluating methods of “in-filling” the removed skates using a GLM or spatial modeling techniques, rather than just leaving them out.** Simply leaving out the skates will only be unbiased if they are a random selection of all skates in a stratum.

A study using data from 1998-2004 suggests that the impact of sperm whale interactions on catch rates is small (~2% for sets with observed depredation). However, there are concerns that the depredation extent is increasing in recent years (in particular, for 8 stations in the W and E Yakutat slope area). The depredation rate is similar to that observed in other fisheries in other parts of the world. Industry views were expressed that the depredation rate is higher than these estimates. Significant changes in the depredation rate will impact both the index of abundance and apportionment schemes. **The Panel supports the proposals to develop better ways to quantify impacts including acoustic techniques, hook monitoring, deterrents, set/skate classification (depredated or not), masking vessel noise, and innovative ways to compare between indices (e.g., parallel pot sets).**

Whale depredation is an issue for fishery CPUE data, as encounters typically lead to vessels leaving an area or in some areas changing to pot gear as they have in the Bering Sea. Quantifying the effect may however be difficult, because the recording of whale depredation incidences in logbooks is incomplete and may not provide a reliable indicator of the true incidence of depredation and its consequences for vessel activity.

We agree. Killer whale depredation, while steady over time, is variable among stations. Simply removing stations greatly increases variability depending on whether it was a high station or low station from the previous years. Sperm whale depredation effects are even more elusive, but certainly could bias the index if they are focused on certain high or low stations that were previously not depredated. We will investigate GLMs to standardize survey and fishery abundance indices and include killer and sperm whale interactions as variables in the model. This is a goal for the 2010 assessment cycle.

3C.6 TOR 5: Recommendations for further improvements

Age and length data. The Panel recommends that comparisons between the length frequency distribution of the age-samples with the overall length frequency be undertaken as an internal consistency check for sampling bias. Furthermore, it would be desirable to develop age-length keys (ALKs) and apply these to the observed length frequency distributions to compare the resulting raised age composition estimates with the randomly sampled age compositions.

We have done this informally in the past and it has yielded similar results for length frequencies of age samples and overall length frequencies. We will use the ALK diagnostic when we are examining a smooth growth regime transition or internally estimated growth.

Spatial structure. An area-disaggregated assessment approach should ideally be developed and may lead to improved management advice. Abundance trends and size/age composition vary by area, and spatially separable index and composition data and movement data from tagging are available. Such a model can also provide better insight on the impact of apportionment policies. Area disaggregation options include:

- o Treating areas as separate fisheries, fitting area-specific selectivity.
- o Modeling movement between areas using tagging information.

A spatially explicit model has its tradeoffs with parsimony. It is well known that spatially-explicit modeling is demanding in terms of the amount of data that needs to be collected. While we agree that the aggregated area model has its limitations in terms of biological realism, it may be more robust in terms of withstanding future changes in the environment or data collections. That being said, it is our goal to evaluate the potential and limits of the scale for which we have sufficient data to provide meaningful results.

Size selectivity. Selectivity is currently modeled by age separately by sex, and the difference in the fitted selectivity curves appear to be largely due to growth differences by sex. **The Panel recommends that size-based selectivity be implemented in future assessments, and that single combined-sex selectivity curves be tested for each fishery.** This will potentially reduce the number of selectivity parameters used by the model.

We have intended to evaluate size-based selectivity for some time. While we have currently been spending time working on sex disaggregating the model, we agree that size based selectivity might be a way to reduce parameters, which will be needed if moving to spatially explicit models.

Sensitivity analyses. Sensitivity of estimated depletion and recommended ABC to important fixed parameters should be part of the assessment documentation.

We intend to be more explicit as to the sensitivity of certain assumptions about key parameters such as natural mortality and catchability in future assessments.

Model building/specification. It would be useful to have a more formal examination of the basis for decision making when building towards the final model configuration and adding individual data sets. Also, the impact of “smoothing” factors (e.g., annual fishing mortality and recruitment) should be evaluated and avoided if unnecessary.

When recommending a new model configuration, we present the addition of each data source or change separately as a possible model. However, one of our goals is to track the progression of the model configurations over time in the assessment document. Current “smoothing” factors have a minimal effect on the model and aid in model convergence.

Growth parameter estimation. Growth parameters should be estimated within the assessment model so that the impact of size-based selectivity is properly accounted for. The sablefish growth parameters have high t_0 values that may be symptomatic of not accounting for selectivity when fitting growth models.

We agree it is worthwhile to attempt to estimate growth within the model, particularly while evaluating size-based selectivity. One disadvantage is that model input files become increasingly cumbersome as thousands of age-length-weight observations are input for every year.

Simulation testing. The current model should be validated by simulation testing using simulated data to ensure that biomass and recruitment trends are reasonably reproduced.

We have done some simulation testing of the model using a simpler operating model with results that support the results of the current model. We will attempt to formalize that work in the future.